Lab Project on Fine-Tuning Language Models

Advanced Natural Language Processing Course

University Year : 2024/2025

Modeling

Choose a dataset from the list below and build a fine-tuned model to perform the task at hand

1. Fine-tune the target model (see table below) on your training dataset and assess the performance on the test dataset
2. Implement a multi-head attention class based on **differential attention** (see appendix)
3. Using your fine-tuned model, replace between 25% to 50% of the encoder and/or decoder layers' multi-head attention with multi-head differential attention. For each configuration you propose, train the new model, and assess the performance

Datasets

NLP Task	Dataset	Dataset Link	Target Model
Text Classification	Arabic Sentiment Tweets Dataset	`https://data.mendeley.com/datasets/57zpx667y9/2`	BERT
Question Answering	ArabicaQA	`https://github.com/DataScienceUIBK/ArabicaQA`	GPT-2
Text Summarization	Essex Arabic Summaries Corpus	`https://www.lancaster.ac.uk/staff/elhaj/corpora.html`	T5

Guidelines

- No written reports are required; only an original PyTorch code accessed via a public GitHub repo link
- Depending on your task, your starting point for fine-tuning could be a model from the list below:
  - Pre-trained BERT model: `https://huggingface.co/aubmindlab/bert-base-arabertv2`
  - Pre-trained GPT-2 model: `https://huggingface.co/aubmindlab/aragpt2-base`
  - Pre-trained T5 model: `https://huggingface.co/UBC-NLP/AraT5v2-base-1024`
- You can choose among the implemented metrics in `torchmetrics` under "Text" tag the ones that best assess your model's performance depending on your task:
  `https://lightning.ai/docs/torchmetrics/stable/all-metrics.html`
- The 10-minute oral presentation (slides are required) should highlight: (1) the goals of your task, (2) the exploratory data analysis, (3) the modeling approach, (4) the design choices of hyperparameters, (5) the final model's performance and evaluation, (6) a live demo on notebook/terminal with a test sample (no UI required), and (7) the potential improvements on your model

Deliverables Deadline

Codes must be submitted before **22/12/2024** at **16:00**
P.S. Submissions will not be accepted after this date

Deliverables Link

`https://forms.office.com/r/EQQyL5Eeui`

Project Defense Date

On **23/12/2024**; further details will be shared via email at a later time
P.S. All group members are required to attend the defense, except in cases of justified absence

Appendix

Reading: `https://arxiv.org/pdf/2410.05258`

Differential attention improves performance by amplifying attention to relevant context while filtering out noise

We denote N the number of tokens, d_{model} the embeddings dimension, d the head dimension of the Transformer, $X \in \mathbb{R}^{N \times d_{\text{model}}}$ the model's input, $Q_1, Q_2, K_1, K_2 \in \mathbb{R}^{N \times d}$ the queries and keys, $V \in \mathbb{R}^{N \times 2d}$ the values

The differential attention operator is defined as follows:

$$
\begin{gathered}
[Q_1; Q_2] = XW^Q, \quad [K_1; K_2] = XW^K, \quad V = XW^V \\
\text{DiffAttn}(X) = \left(\text{softmax}\left(\frac{Q_1 K_1^T}{\sqrt{d}} \right) - \lambda \, \text{softmax}\left(\frac{Q_2 K_2^T}{\sqrt{d}} \right) \right) V
\end{gathered} \tag{1}
$$

where $W^Q, W^K, W^V \in \mathbb{R}^{d_{\text{model}} \times 2d}$ are parameters, and λ is a learnable scalar defined as follows:

$$
\lambda = \exp(\lambda_{\mathbf{q_1}} \cdot \lambda_{\mathbf{k_1}}) - \exp(\lambda_{\mathbf{q_2}} \cdot \lambda_{\mathbf{k_2}}) + \lambda_{\text{init}} \tag{2}
$$

where $\lambda_{\mathbf{q_1}}, \lambda_{\mathbf{k_1}}, \lambda_{\mathbf{q_2}}, \lambda_{\mathbf{k_2}} \in \mathbb{R}^d$ are learnable vectors, and $\lambda_{\text{init}} \in (0, 1)$.
You can set $\lambda_{\text{init}} = 0.8$ and the number of heads $h = d_{\text{model}}/2d$.

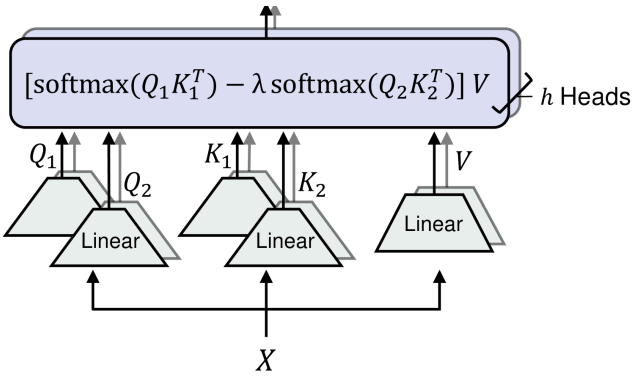

Figure 1: Multi-head differential attention